A-A-2537
March 25, 1987

COMMERCIAL ITEM DESCRIPTION

TRAMMELS

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-T-661.

This commercial item description covers trammels to be used by craftsmen for laying out work and for measuring distances between points too great for ordinary dividers.

Salient characteristics:

Trammels assigned National Stock Number (NSN) 5210-00-221-1848 shall be of a precision type and consist essentially of a pair of trammel heads, fine adjustment attachment, a pair of scriber points, a pair of caliper legs, a steel beam, and shall be similar to figure 1.

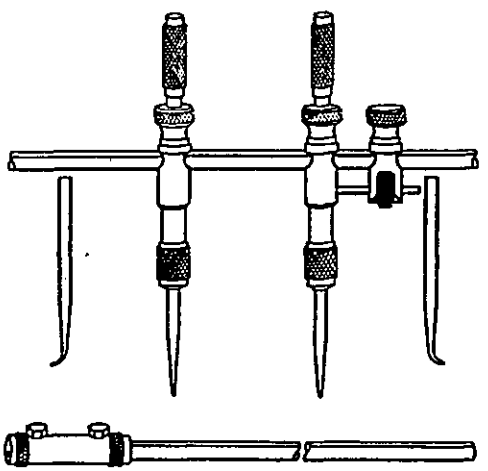


Figure 1. Extension steel beam trammels, fine adjustment.

Trammel heads. One head shall consist of a shaft having a hole to fit neatly on the steel beam and a chuck for holding points. A locking nut with a knurled grip shall be provided for holding the head securely on the beam. A spring friction device shall be provided to hold the head in position when the nut is loosened. The other head shall be similar, except that a fine adjustment clamp and adjusting screw shall be attached.

Chucks. Chucks shall be of such design as to grip and firmly hold the points at any position within the adjustment range. The chucks shall be knurled on the exterior surface.

Locking nuts. Locking nuts shall be knurled on the periphery.

Fine adjustment clamp. The fine adjustment clamp shall consist of a yoke to accommodate a fine adjustment screw and nut. The clamp shall be made to fit neatly on the steel beam, and be provided with a locking screw for holding the clamp firmly in position. Clearance between the yoke and bearing surface of nut shall not exceed 0.005 inch.

FSC 5210

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Fine adjustment screw and nut. One end of the screw shall pass through the yoke of the fine adjustment clamp and the nut located in the yoke. The other end of the screw shall be attached to one of the heads. The screw shall provide a fine adjustment of not less than 3/8 inch. The nut shall be circular in shape and knurled on the periphery.

Points. Trammels shall be furnished with two replaceable scriber points which shall be of a good tool steel, hardened and ground. They shall have a straight shaft with one end uniformly tapered for approximately 1 inch to a sharp point. Points shall be not less than 3 nor more than 5 inches in length.

Caliper legs. Caliper legs shall conform to the requirements for scriber points except that the pointed section shall be shaped as the legs of inside calipers.

Beams. Beams shall be of steel rod, with a flat to prevent heads from turning when clamped in position. The beam shall be 5/16 inch, plus or minus 1/64 inch, in diameter; 20 inches, plus or minus 1/8 inch, in length; and shall consist of one section.

Finish. The finish shall be natural polish, nickel plate, chrome plate, black oxide, or black wrinkle.

Screw threads shall be of fine pitch or lead in accordance with FED-STD-H28.

Trammels shall be furnished in a wooden case.

Trammels assigned National Stock Number (NSN) 5210-00-221-1850 shall consist essentially of two heads with removable scriber points, locking screws, a fine adjustment device, a pencil attachment, and shall be similar to figure 2.

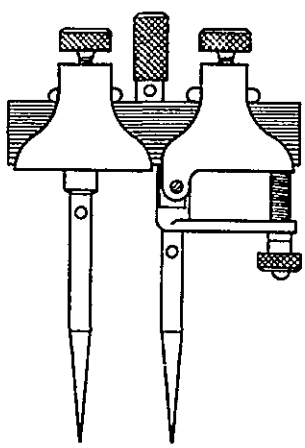


Figure 2. Trammels without beam, fine adjustable points.

Heads. Heads shall be of bronze, brass, zinc alloy die casting, or steel. They shall be suitable for attaching to a wooden beam 3/4 inch wide and 5/16 inch thick, shall have a locking screw, and suitable means for attaching points. One head shall have a fine adjustment attachment.

Locking screws. Locking screws shall be a good fit in the threaded hole of the head. The outer end of the screw shall have a circular knurled head. The inner end shall have a metal block or plate attached to contact the beam and hold the head firmly in place. The block or plate shall not turn with the screw.

Fine adjustment attachment. A fine adjustment device shall be attached to one of the trammel heads in such a manner as to provide a fine adjustment of at least 1/2 inch.

Points. Points shall be of a good grade tool steel, hardened and ground. They shall have a straight shaft with one end uniformly tapered for approximately 1 inch to a sharp point. They may be threaded for attaching to the head. Points shall be not less than 2-1/2 nor more than 5 inches long.

Finish. Finish shall be either natural polish, nickel plate, chrome plate, black oxide, or black wrinkle.

Screw threads shall be of fine pitch or lead in accordance with FED-STD-H28.

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Workmanship. Workmanship shall be first class throughout. The trammels shall be free from defects which may affect serviceability or durability. Burrs and sharp edges shall be removed.

Marking. Each trammel shall be marked in a plain and permanent manner with the manufacturer's name or identifying symbol, part number, and country of origin.

Quality assurance provisions.

Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes, except where otherwise indicated herein. The lot size shall consist of all trammels offered for acceptance at one time. The sample unit is one trammel.

Examination for visual and dimensional requirements. Each trammel shall be examined for workmanship, marking, and dimensional requirements. The inspection level shall be Inspection Level S-1 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Testing for scriber point hardness. Each scriber point shall be examined for a hardness equivalent to 50 to 56 on the Rockwell "C" scale when tested. The end of the points may be ground off a sufficient amount to allow for taking of the Rockwell hardness readings. The points shall be examined with a Special Inspection Level of S-1 with an AQL of 4.0 defects per hundred units.

Regulatory requirements. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order. (For use by GSA; optional for other agencies.) DoD may use Commercial Packaging, Standard ASTM D3951.

NOTES:
The issue of referenced documents in effect on the date of the solicitation shall be used to determine compliance with stated requirements.

Federal Standards are available from General Services Administration Specifications Unit (WFSIS), 7th and D Streets, SW, Washington, DC 20407.

Military Standards are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

ASTM Standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

MILITARY INTERESTS:

Custodian

Air Force - 99

Review Activity

Air Force - 84

PREPARING ACTIVITY:

GSA/FSS

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